

Exercise 1: Mocking and Stubbing with Mockito

Testing a service that depends on an external API, using JUnit 5 + Mockito

1. pom.xml (relevant dependencies & build)

pom.xml

```
<dependencies>
  <dependency>
    <groupId>org.junit.jupiter</groupId>
    <artifactId>junit-jupiter</artifactId>
    <version>5.10.2</version>
    <scope>test</scope>
  </dependency>

  <dependency>
    <groupId>org.mockito</groupId>
    <artifactId>mockito-core</artifactId>
    <version>5.17.0</version>
    <scope>test</scope>
  </dependency>
</dependencies>

<build>
  <plugins>
    <plugin>
      <groupId>org.apache.maven.plugins</groupId>
      <artifactId>maven-surefire-plugin</artifactId>
      <version>3.2.5</version>
    </plugin>
  </plugins>
</build>
```

2. ExternalApi.java (dependency to be mocked)

src/main/java/ExternalApi.java

```
public interface ExternalApi {
    String getData();
}
```

3. MyService.java (class under test)

src/main/java/MyService.java

```
public class MyService {

    private final ExternalApi externalApi;

    public MyService(ExternalApi externalApi) {
        this.externalApi = externalApi;
    }

    public String fetchData() {
        return externalApi.getData();
    }

}
```

4. MyServiceTest.java (mocking & stubbing test)

src/test/java/MyServiceTest.java

```
import static org.junit.jupiter.api.Assertions.assertEquals;
import static org.mockito.Mockito.*;

import org.junit.jupiter.api.Test;

public class MyServiceTest {

    @Test
    public void testExternalApi() {
        // 1. Create a mock object for the external API
        ExternalApi mockApi = mock(ExternalApi.class);

        // 2. Stub the method to return a predefined value
        when(mockApi.getData()).thenReturn("Mock Data");

        // 3. Use the mock in the service under test
        MyService service = new MyService(mockApi);
        String result = service.fetchData();

        assertEquals("Mock Data", result);
    }

}
```